Efficacy of heavy eccentric calf training for treating mid-portion Achilles tendinopathy: a systematic review and meta-analysis

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ABSTRACT

Objective To assess the effectiveness of heavy eccentric calf training (HECT) in comparison with natural history, traditional physiotherapy, sham interventions or other exercise interventions for improvements in pain and function in mid-portion Achilles tendinopathy.

Design A systematic review and meta-analysis were conducted as per the PRISMA guidelines.

Data sources PUBMED, CINAHL (Ovid) and CINAHL (EBSCO) were searched from inception until 24 September 2018.

Eligibility criteria Randomised controlled trials comparing HECT to natural history, sham exercise, traditional physiotherapy and other exercise interventions were included. Primary outcome assessing pain and function was the Victorian Institute of Sports Assessment-Achilles.

Results Seven studies met the inclusion criteria. This review suggests HECT may be superior to both natural history, mean difference (MD) (95% CI) of 20.6 (11.7 to 29.5, one study) and traditional physiotherapy, MD (95% CI) of 17.70 (3.75 to 31.66, two studies). Following removal of one study, at high risk of bias, due to pre-planned sensitivity analysis, this review suggests HECT may be inferior to other exercise interventions, MD (95% CI) of −5.65 (-10.51 to −0.79, three studies). However, this difference is unlikely to be clinically significant.

Conclusion Current evidence suggests that HECT may be superior to natural history and traditional physiotherapy while HECT may be inferior to other exercise interventions. However, due to methodological limitations, small sample size and a lack of data we are unable to be confident in the results of the estimate of the effect, as the true effect is likely to be substantially different.

Systematic review registry PROSPERO registration number: CRD4201804493

Protocol reference This protocol has been published open access: Murphy M, Travers MJ, Gibson, W. Is heavy eccentric calf training superior to natural history, sham rehabilitation, traditional physiotherapy and other exercise interventions for pain and function in mid-portion Achilles tendinopathy? *Systematic Reviews* 2018; 7: 58

Summary

What is already known

- ► Heavy eccentric calf training, also known as the Alfredson eccentric protocol, is commonly used in clinical practice for mid-portion Achilles tendinopathy.

What are the new findings

- ► Heavy eccentric calf training may be better than wait-and-see and traditional physiotherapy for mid-portion Achilles tendinopathy but the quality of the evidence is very low and low, respectively.
- ► Other exercise interventions, such as heavy slow resistance training may be superior to heavy eccentric calf training for mid-portion Achilles tendinopathy, however the quality of the evidence is low and this difference is unlikely to be clinically significant.
- ► The findings of this review challenge current clinical practice and would suggest that the given evidence supporting the use of heavy eccentric calf training versus natural history is very low. Further research with more robust studies are needed.

BACKGROUND

Description of the condition

Achilles tendinopathy (AT) is a common musculoskeletal disorder. Epidemiological data indicates an incidence of 2.16 per 1000 patient years, in the general population, and a prevalence of 6.2%–9.5% running-related injuries in athletes.[1 2] Clinically, tendinopathy is characterised by focal pain, morning stiffness and restricted function.[3]

Description of the intervention

Exercise-based interventions have been suggested to be most effective in the management of mid-portion AT,[4] where a meta-analysis of 24 studies showed that a mean improvement of 21 points on the Victorian Institute of Sports Assessment – Achilles (VISA-A) was observed at 12 weeks.[5] To date, six different exercise interventions for AT have been reported: heavy eccentric calf training (HECT) which is also referred to as the Alfredson eccentric protocol;[6] modified HECT that is a low-volume version of HECT;[7] concentric training;[8] eccentric overload training that is also referred to as the Silbernagel protocol;[9] heavy slow resistance training;[10] and the 'Stanish protocol'.[11] The differences between these protocol are described in online supplementary appendix A. A recent narrative review suggested that no protocol is superior to another when comparing outcomes in pain and

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function.[4] However, this comparison has yet to be examined under systematic review methodology with a systematic search strategy, grading of the body of the evidence and meta-analysis. Likewise, statistical comparisons of the differing exercise interventions against a natural history group (as assessed via wait-and-see) or sham rehabilitation has not been made. A 2009 systematic review compared HECT versus any other intervention for treating mid-portion AT with only three studies included and no pooled data synthesis.[12] A systematic review from 2015 investigating different eccentric exercise interventions for mid-portion AT found no difference between the traditional HECT intervention and lower load versions of HECT: again, no pooled data synthesis was performed.[13] Both of these systematic reviews were unable to make firm conclusions due to poor reporting of compliance and significant heterogeneity of both the study population and outcome measures.[12 13] This prevented these reviews using meta-analysis. Of note, neither of these systematic reviews compared HECT to either a natural history or sham rehabilitation group. When trying to establish the efficacy of an intervention this is an essential component of the systematic review.[14]

Why is it important to do this review?

Exercise rehabilitation, specifically HECT, is commonly used in clinical practice for rehabilitation of mid-portion AT.[12 13] However, this approach has yet to be systematically compared with natural history, sham intervention or traditional physiotherapy groups. Thus, estimates of the efficacy/effectiveness of HECT have not been established.

This review has clinical value. HECT is more time-consuming, with 180 repetitions per day,[6] compared with other exercise interventions, such as heavy slow resistance training which only requires 18–60 repetitions three times per week.[10] HECT is also difficult to integrate with other aspects of rehabilitation such as running or plyometric training, as current dosage recommendations are that it must be completed at least once-daily with no scheduled rest days.[6] As such, we seek to estimate whether other exercise interventions afford similar outcomes as has been previously suggested.[4] If less time-consuming exercise interventions are equally effective for rehabilitation of mid-portion AT it may also have implications on adherence to treatment[10] and the ability to engage in other aspects of a comprehensive management plan such as plyometric training and functional rehabilitation (eg, running).

Objectives

To assess the effectiveness of heavy eccentric calf training in comparison to natural history, traditional physiotherapy, sham interventions or other exercise interventions for improvements in pain and function in mid-portion.

METHODS

Systematic review registration

The protocol for this systematic review was registered on the PROSPERO: CRD4201804493 and has been published open access: Murphy M, Travers MJ, Gibson, W. Is heavy eccentric calf training superior to natural history, sham rehabilitation, traditional physiotherapy and other exercise interventions for pain and function in mid-portion Achilles tendinopathy? *Systematic Reviews* 2018; 7: 58.[15]

Deviations to protocol

The original protocol was submitted to PROSPERO on 18 December 2017 and was modified on the 5 March 2018 to incorporate recommendations made by reviewers at *Systematic Reviews* during the protocol publication process[15] to include more in-depth assessment of heterogeneity and subsequent sensitivity analysis. Data extraction, risk of bias assessment and data analysis did not occur until publication of the protocol in *Systematic Reviews* on 13 April 2018. The PROSPERO record was further updated on 1 June 2018 to update the progress of the review.

Data management

Data was managed and stored via The University of Notre Dame Australia's online file server and onedrive.

Criteria for considering studies for this review

Types of studies

Randomised and quasi-randomised trials were included if one study arm used HECT to treat mid-portion AT and the other used a natural history, sham rehabilitation, traditional physiotherapy or other exercise intervention. Studies that used an exercise intervention with a control or sham/placebo intervention were included (eg, one arm of the study used HECT with a sham intervention and another arm used a modified version of HECT with a sham intervention). Identified studies were included regardless of their publication status. Trials that were non-randomised observational trials, case reports/series, clinical observations and systematic reviews were excluded.

Types of participants

Both physically active and sedentary participants aged 18 years and over identified as having mid-portion AT for greater than 3 months were included. Studies that included participants with insertion AT or other causes of heel pain were excluded from the review.

Types of natural history and sham interventions

Natural history groups were included if they had no form of active or passive intervention and were treated as a wait-and-see group. Sham intervention groups were classified by the reviewers (MM/MT/WG) as sham if the participants underwent sham exercise interventions highly unlikely to result in a pathophysiological response which would influence participant pain and function, apart from a placebo response. Specifically, exercise interventions that do not overload contractile tissue and are physiologically unlikely to induce a strength response were classified as sham rehabilitation.

Types of traditional physiotherapy

Traditional physiotherapy groups were included if the participants had no exercise intervention but had some form of traditional physiotherapeutic intervention as defined by Weidmann *et al*.[16] The traditional physiotherapy interventions included were:

- Deep friction massage to the tendon, and/or
- Other forms of manual physical therapy to local tissues, and/or
- Ultrasound, and/or
- Taping.

There is a lack of evidence regarding the effectiveness of these interventions.[17] However, there is also no clear evidence to suggest whether these interventions may or may not interfere with an exercise intervention. Therefore, we made the conservative decision to classify the interventions above as traditional physiotherapy and not sham interventions.

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Other non-invasive interventions such as shockwave therapy were excluded as the efficacy of shockwave therapy has been examined in multiple reviews[18–20] and does not conform to 'traditional physiotherapy'.[16 21–23]

Types of interventions

Exercise intervention studies were included if they used either isometric, eccentric, concentric or isotonic (eccentric and concentric) exercise interventions. Studies were also included if they had an isometric, eccentric, concentric or isotonic exercise intervention in conjunction with a placebo therapy (for example, sham laser treatment) as long as the comparator group within the same study underwent an identical placebo therapy.

Comparisons of interest were:

- HECT versus natural history.
- HECT versus sham exercise.
- HECT versus traditional physiotherapy.
- HECT versus different exercise interventions.

Primary outcome measure

It has been reported in a previous systematic review comparing different eccentric interventions for mid-portion AT that the heterogeneity of outcome measures led to an inability to report conclusive results.[13] Therefore, only studies that used a validated and reliable outcome measure of pain and function in mid-portion AT were included. The VISA-A is a self-reported outcome measure which includes questions on Achilles tendon pain, stiffness and functional capacity.[24] The VISA-A is a valid ($P<0.01$) and reliable (test-retest reliability $r=0.98$) tool[24] that has been recognised as the only valid and reliable measure to assess pain and function in AT.[25]

Trials which used the visual analogue scale (VAS) or numerical rating scale of pain were not included. Besides the VAS having poor test-retest reliability for mid-portion AT ($r=0.45$,[9] it also fails to recognise the influence of load. The relationship between pain and function is intricately linked in tendinopathy given symptoms are load-dependent,[3 26] so including a measure of pain without linking it to function may result in an inaccurate estimate of the effect.

Search methods for identification of studies

Searches were conducted from inception until 31 December 2017 and updated on 24 September 2018 to ensure no additional records were eligible for inclusion.

Electronic searches

Searches were conducted using free text terms (table 1) to identify published articles on the following electronic databases:

- PUBMED.
- CINAHL (Ovid).
- CINAHL (EBSCO).

Only peer-reviewed, human, clinical trials were included, however this was adapted to individual databases as necessary. We did not restrict the language of publication to English.

Table 1 Systematic review search strategy

Number	Combiners	Terms
1	Problem of interest	Achil* OR triceps surae* OR tend* OR heel OR calcan*
2	Intervention	Exercise OR eccentric* OR isotonic* OR heavy slow resistance OR isometrics OR resistance OR strength* OR Alfredson*
3		#1 AND #2
4	Outcome	VISA* OR Victorian Institute of Sport score*
5		#3 AND #4
	Limitations	Peer-reviewed, human, clinical trials

Searching other resources

Additional searches were conducted on the Cochrane Central Register of Controlled Trials, metaRegister of controlled trials (mRCT) (www.controlled-trials.com/mrct), clinicaltrials.gov (www.clinicaltrials.gov), and the WHO International Clinical Trials Registry Platform (ICTRP) (apps.who.int/trialsearch/) for ongoing trials.

Reference lists for reviews and retrieved articles for additional studies were checked and citation searches on key articles performed. Experts in the field were contacted for unpublished and ongoing trials. The list of included studies were evaluated by content experts to help identify any additional relevant studies. Web of Science was also used for forward citation tracking.

The ePublication lists of key journals in the field were screened in an attempt to pick up studies which had yet to be indexed in the databases. The journals searched are listed in full within our protocol.[15]

Unpublished data

To minimise the prospect of publication bias, a further search of the following databases was undertaken:

- OpenGrey (System for Information on Grey Literature in Europe).
- Dissertation Abstracts (Proquest).
- SPORTDiscus.

Data collection and analysis

Selection of studies

Two review authors (MM and MT) independently assessed the titles and abstracts of potential studies identified by the search strategy for their eligibility. If the eligibility of a study was unclear from the title and abstract, the full article was then assessed. Studies that did not match the inclusion criteria for this review were excluded. Studies were not anonymised prior to assessment.

A PRISMA study flow diagram has been included in the full review to document the screening process.[27]

Data extraction and management

Two review authors (MM and MT) independently extracted data from all included studies using a standardised and piloted data extraction form. The following information was extracted from included studies: primary author, year of publication, study design, sample size, baseline demographics, intervention, adherence, concomitant treatments, mean (SD) of the VISA-A at baseline and all follow-up time points during the intervention period, time (weeks) at each follow-up point.

Assessment of risk of bias in included studies

Two review authors (MM and MT) assessed risk of bias for each study independently.

Included studies were assessed using the Cochrane Risk Of Bias (RoB 2.0) tool.[28] The RoB 2.0 tool involves judgement on seven domains as described by the Cochrane Colloquium (http://www.riskofbias.info).[28] Judgements on the risk of bias for each of the domains and overall risk of bias were made as per the recommendations of the RoB 2.0 tool. Using the RoB 2.0 tool all

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studies were classified overall as having either a low risk of bias, some concerns of bias or high risk of bias.

Measure of treatment effect

The VISA-A was used across all trials and therefore, all primary outcomes were presented and analysed on a continuous scale as mean difference (MD) with 95% confidence intervals (95% CI). The final time point at which a study assessed participants' VISA-A while they were still completing the intervention of interest was used for analysis.

Dealing with missing data

Study authors were contacted where insufficient data was presented in the study report to enter into meta-analysis. If a study did not provide data on the sample size and mean and these data were not provided following contact with the author, the study was excluded from the meta-analyses.

Where the SD of the study was not available in the study report or after contact with the authors the SD was inputted from another trial that used the same outcome measure (VISA-A) at an identical follow-up time point. The cohort selected to extract the SD which was used for the cohort missing the SD was randomly chosen provided from the other included studies which had an identical follow-up time point. This method is recommended in Part 3, Section 16.1.3.2 of the *Cochrane Handbook for Systematic Reviews of Interventions*.[28]

Assessment of heterogeneity

Unlike previous studies which had significant heterogeneity due to participants and outcome measures, we restricted our review to mid-portion which had been assessed using the only valid and reliable outcome measure for mid-portion AT, the VISA-A. Therefore, given the strictly defined inclusion criteria for studies (diagnosis/condition, exercise intervention and outcome measure), we anticipated clinical heterogeneity would be limited.

A Chi square test evaluated the statistical significance of heterogeneity. The I^2 statistic estimated the amount of study heterogeneity based on the P-value being <0.10 or the I^2 value being $>40\%$,indicating significant heterogeneity.[28]

Where significant heterogeneity was found, a sub-group analysis investigating the impact of a study was determined by completing a sensitivity analysis. This analysis involved exclusion of pre-determined sub-groups from heterogeneity analysis. Using the aforementioned statistical tests, the heterogeneity of the remaining studies was determined.

Sensitivity analysis

A sensitivity analysis was conducted by allocating a weight of 0% to different sub-groups to assess their influence on the overall effect size and measures of heterogeneity. A pre-planned[15] sensitivity analysis was conducted in the following subgroups:

- Studies in which the SD was inputted as per the Methods section above.
- Studies in which the adherence was not reported.
- Different exercise interventions compared against HECT.
- Studies in which both HECT and the exercise intervention used as the comparator both received placebo interventions.
- Studies in which there was a high risk of bias as assessed by the RoB 2.0 tool.

Assessment of reporting biases

The risk of bias criterion 'study size' was used to address the potential influence of small study biases.[29] If a study contained a sample size of less than 50 participants, it was considered at high risk of small sample bias. If a study contained between 50 and 200 participants, it was considered at moderate risk of small sample bias. If a study contained greater than 200 participants, it was considered at low risk of small sample bias.

The likelihood of reporting biases were to be investigated when more than 10 studies were included in the meta-analysis using funnel plots representing each follow-up time point, provided sample sizes were not identical.[30] For continuous outcomes the Egger's test[31] was to be used to detect possible small study bias.[28] However, none of the meta-analysis groups had greater than 10 studies and therefore neither of these methods were completed.

Assessment of the quality of the body of evidence

Assessment of the quality of the body of evidence was assessed using the GRADE approach[32] as recommended in Part 2, Section 12.2.1 of the *Cochrane Handbook for Systematic Reviews of Interventions*.[28] The GRADE approach involves making an overall judgement on the quality of the body of evidence based on the overall risk of bias, consistency of results, directness of the evidence and publication bias.[32]

Data synthesis

The data were analysed using an inverse variance and random effects analysis model in Review Manager version 5.3 to calculate the mean difference (MD) (+/-95% CI).[33]

RESULTS

Selection of studies

There were no disagreements between the two review authors on study inclusion. A total of seven studies met the criteria and were included in these meta-analyses (figure 1). Six studies were written in English[7 10 21 34–36] and one in German which was translated into English by a native German speaker prior to data

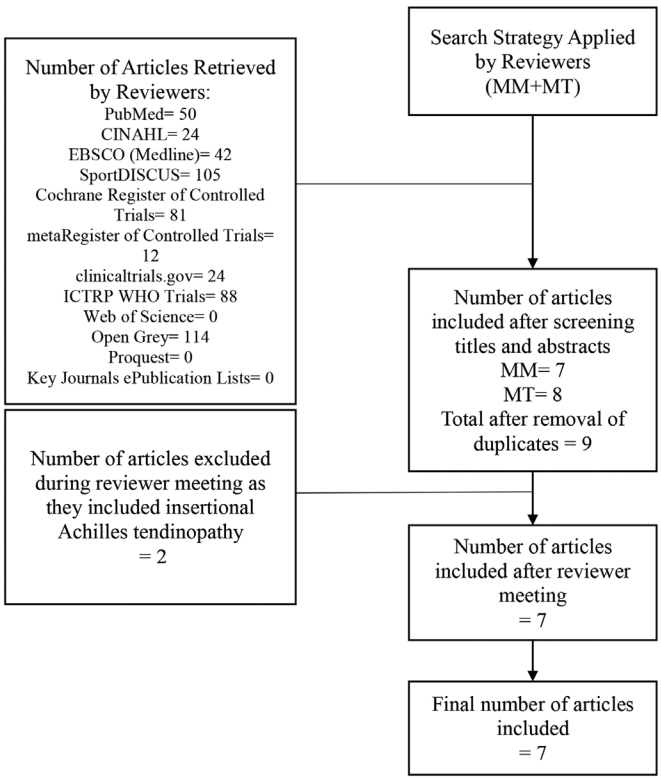

Figure 1 PRISMA study flow diagram.

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Table 2 Individual study characteristics

Study name	Study design	Sample size (n)	Mean age (SD)	Symptom duration in months (SD)	Male/female	Intervention	Adherence recorded	Follow-up time point (weeks)	Baseline mean VISA-A (SD)	Final mean VISA-A (SD)
Beyer et al[10]	RCT	25	48 (10)	19 (25)	18/7	Heavy eccentric calf training	78%	12	58 (19.5)	72 (18.5)
		22	48 (9.38)	17 (15)	14/8	Heavy slow resistance training	92%	12	54 (15)	76 (18.5)
Herrington and McCulloch [21]	RCT	13	37 (9.26)	21.3 (17.7)	Not reported	Heavy eccentric calf training	Not reported	12	62 (18)*	97 (15)*
		12	36.6 (7.14)	27.6 (12.9)	Not reported	Traditional physiotherapy (deep friction massage and ultrasound)	Not reported	12	60 (20)*	82 (30)*
Rompe et al [36]	RCT	25	48.1 (9.9)	10.9 (7.7)	9/16	Heavy eccentric calf training	Not recorded	16	50.6 (11.5)	75.6 (18.7)
		25	46.4 (11.4)	9.2 (10.5)	9/16	Natural history	Not recorded	16	48.2 (9.0)	55 (12.9)
Stasinopoulos and Manias [35]	RCT	20	48.2 (5.1)	7.1 (not reported)	Not reported	Heavy eccentric calf training	Not reported	12	36 (23.4)	76 (18.5)
		21	48.4 (5.1)	6.9 (not reported)	Not reported	Stanish Protocol	Not reported	12	38 (23.4)	63 (18.5)
Stevens and Tan [7]	RCT	14	48.2 (10.8)	6.2 (2.1)	6/9	Heavy eccentric calf training	>75%	6	49.6 (10.2)	58.7 (13)
		12	49.2 (11.3)	8.9 (5.1)	5/8	Modified heavy eccentric calf training	>75%	6	47.1 (15.6)	62.5 (12.8)
Tumilty et al[34]	RCT	13	47.2 (8.5)	Not reported	9/11	Heavy eccentric calf training	70%	12	56.7 (12)	80.4 (9.66)
		19	47.7 (10.1)	Not reported	7/13	Modified heavy eccentric calf training	100%	12	56.8 (11.3)	87.6 (9.11)
Wiedmann et al[16]	RCT	10	43 (6)	Not reported	7/13	Heavy eccentric calf training	Not reported	12	56 (18)	72 (15)
		10		Not reported		Traditional physiotherapy (deep friction massage and ultrasound)	Not reported	12	49 (20)	52 (30)

*The SD for this study was inputted from another trial.

extraction.[16] One study protocol also met the inclusion criteria for the review but given no data was available, was excluded from meta-analysis and reported as ongoing.[37]

Data extraction

There were two discrepancies in the data extraction due to errors that were resolved between the two review authors. The complete datasheet is freely available online at figshare (https://figshare.com/articles/Efficacy_of_heavy_eccentric_calf_training_for_treating_mid-portion_Achilles_tendinopathy_a_systematic_review_and_meta-analysis/7562435). A summary of studies is presented in table 2.

One study compared HECT to a natural history group.[36] No studies compared HECT to a sham/placebo exercise intervention. Two studies compared HECT to traditional physiotherapy.[16 21] Four different studies compared HECT to other exercise interventions.[7 10 34 35]

All studies provided the mean of the VISA-A at both baseline and the final follow-up. However, one study provided the mean but did not report the SD of the group at the final follow-up.[21] The authors were contacted in an attempt to capture the required information but the authors were unable to supply these data.[21] The SD was therefore inputted from another study randomly selected, which was in the same classification group (traditional physiotherapy) and had the same time period of follow-up, as per our protocol.[15] One study provided 95% CI, however the reported 95% CI were not symmetrical around the mean (e.g. mean (95% CI)=76 (77–90)).[35] The authors of this study were contacted for clarification, but no response was received.[35] In this case, the SD was inputted from another randomly selected study that was in the same classification group (other exercise interventions) and had the same time period of follow-up, as per our protocol.[15]

Demographics of the participants

In total the seven studies included 241 participants with a mean (SD) age ranging from 36.6 (7.2) to 49.2 years (11.3). Five of the trials reported the gender spread with the participants being 45% males and 54% females.[7 10 16 34 36] Four of the trials reported body mass index (BMI) with the mean BMI ranging from 25.0 (5.0) to 31.6 (6.1)kg/m^2.[7 10 21 34] Five of the trials reported the duration of symptoms with the mean duration ranging from 6.2 (2.1) to 27.6 (12.9) months.[7 10 21 35 36] All studies reported a baseline VISA-A, with the mean VISA-A at baseline ranging from 36 (23.4) to 62 (18). The duration of follow-up to the point of ending the intervention ranged from 6 weeks to 16 weeks.

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Table 3 Risk of bias assessment for randomised controlled trials using RoB 2.0 tool

Study name	Bias arising from the randomisation process	Bias due to deviations from intended interventions	Bias due to missing outcome data	Bias in measurement of the outcome	Bias in selection of the reported result	Overall
Beyer et al [10]	Low	Low	Low	Low	Low	Low
Herrington and McCulloch et al[21]	Low	Low	Low	Low	Low	Low
Rompe et al[36]	Low	Low	Low	High	Low	High
Stasinopoulos and Manias [35]	High	Low	Low	Low	Low	High
Stevens and Tan[7]	Low	Low	Low	Low	Low	Low
Tumilty et al[34]	Low	Low	Low	Low	Low	Low
Weidmann et al. [19]	Low	Low	Some concerns	Low	Low	Some concerns

Three studies diagnosed as mid-portion AT using a combination of clinical examination and sonography[10 16 36] whereas four studies diagnosed as mid-portion AT using clinical examination alone.[7 21 34 35]

Assessment of heterogeneity

An assessment of heterogeneity was not considered for the comparison of HECT to natural history, given that only one study was included.[36] There was minimal heterogeneity in the studies comparing HECT to traditional physical therapy (Tau^2=0.00, Chi^2=0.18, df=1 (P=0.68) and I^2=0%).[16 21] There was evidence of significant heterogeneity in the studies comparing HECT to other exercise interventions (Tau^2=46.61, Chi^2=9.24, df=3 (P=0.03) and I^2=68%).[7 10 34 35]

Assessment of risk of bias in included studies

Risk of bias was assessed and is presented in table 3. There were no disagreements between review authors. Based on the RoB 2.0 tool the one study which compared HECT to a natural history group is at high risk of bias.[36] One study which compared HECT to traditional physiotherapy has some concerns for risk of bias.[16] Finally, one study which compared HECT to other exercise interventions is at high risk of bias.[35] All other studies were at low risk of bias.[7 10 21 34]

Assessment of reporting biases

All studies had a sample size of 50 or less and were classified as having a high chance of small study bias.[7 10 16 21 34–36] A funnel plot was not graphed for any of the comparisons of interest given there were less than 10 studies in each comparison.[30]

Effect of intervention

HECT versus natural history

One study compared HECT versus natural history with a statistically significant MD (+/-95%CI) of 20.6 (11.69 to 29.51) in favour of HECT.[36] However, the single trial comparing HECT versus natural history[36] was overall at high risk of bias and was at high risk of small sample bias. With consideration of the components of the GRADE system, we rated the overall quality of the body of evidence as very low due to the small sample size, methodological limitations and lack of data. Due to the quality of the evidence being very low we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of the effect.[38]

HECT versus traditional physiotherapy

Two studies were pooled for the comparison of HECT versus traditional physiotherapy.[16 21] There was a statistically significant pooled MD (+/-95%CI) of 17.70 (3.75 to 31.66) in favour of HECT

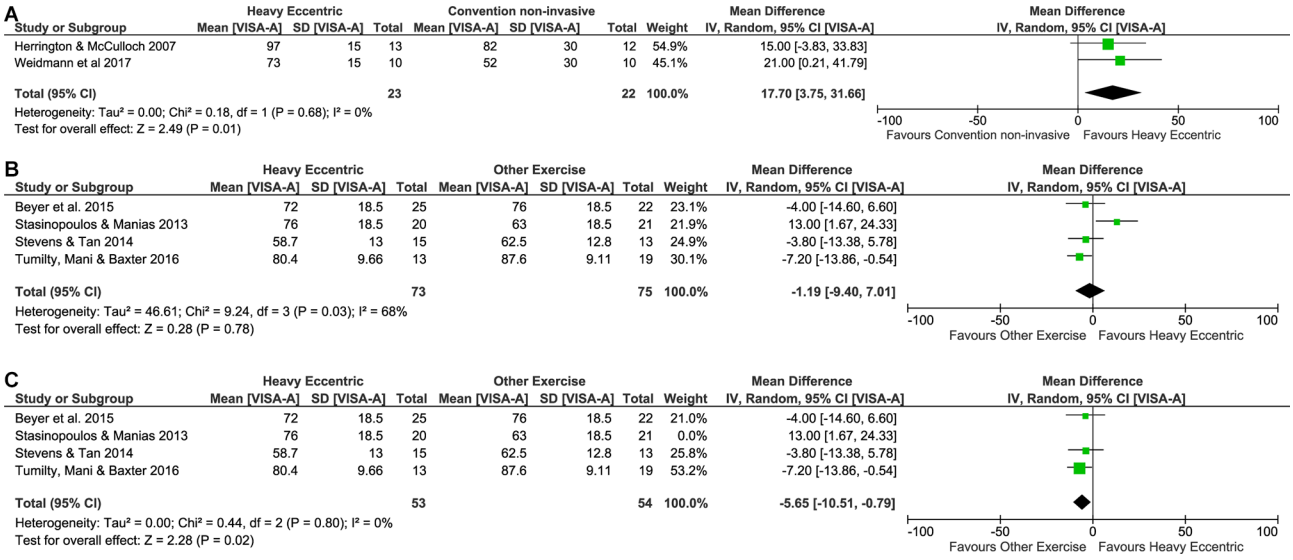

Figure 2 (A) Heavy eccentric calf training versus traditional physiotherapy. (B) Heavy eccentric calf training versus other exercise interventions.(C) Heavy eccentric calf training versus other exercise interventions after removal of one study for sensitivity analysis.

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(figure 2A). Of the two randomised controlled trials included one[16] had some concerns for risk of bias. Both studies were also at high risk for small sample bias (n<50).[16 21] Given the limited number of studies, publication bias could not be assessed. The two studies did not appear to demonstrate inconsistency or indirectness.[16 21] With consideration of the components of the GRADE system, we rated the overall quality of the body of evidence as low due to the small sample size, concern of risk of bias and lack of data. Due to the quality of the evidence being low our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.[38]

HECT versus other exercise interventions

Four studies were pooled for the comparison of HECT versus traditional physiotherapy. Two studies compared HECT to modified HECT[7 34] one study compared HECT to heavy slow resistance training[10] and one study compared HECT to the Stanish Protocol.[35] There was a non-significant pooled MD (+/-95% CI) of −1.19 (-9.40 to 7.01) between exercise interventions (figure 2B). Of the four randomised controlled trials included, one was at high risk of bias.[35] All studies were at high risk for small sample bias (n<50).[7 10 34 35] Given the limited number of studies, publication bias could not be assessed. One study demonstrated a problem with consistency (MD=13.00)[35] and did not support the other three studies (MD=−3.8 to −7.20).[7 10 34] The studies did not appear to demonstrate indirectness.[7 10 34 35] With consideration of the components of the GRADE system, we rated the overall quality of the body of evidence as very low due to risk of bias, small sample sizes and inconsistency. Due to the quality of the evidence being very low, we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of the effect.[38]

Sensitivity analysis

A pre-planned sensitivity analysis was conducted according to our protocol[15] on the comparison of HECT versus other exercise interventions by excluding the one study[35] that had the SD inputted, did not report adherence and was at a high risk of bias. By excluding this study from meta-analysis there was no significant differences in the heterogeneity of the studies ($Tau^2=0.00$, $Chi^2=0.44$, df=2 (P=0.80) and $I^2=0\%$). After excluding this study from analysis, the pooled MD (+/-95% CI) changed from −1.19 (-9.40 to 7.01) to −5.65 (−10.51 to −0.79), respectively (figure 2C). After the removal of the study[35] the results changed from no between-group differences to a significant result in favour of other exercise interventions versus HECT. However, with consideration of the components of the GRADE system, we rated the overall quality of the body of evidence as low due to small sample sizes and lack of data. Due to the quality of the evidence being low our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.[38]

DISCUSSION

Eccentric calf training is commonly used in the management of mid-portion Achilles tendinopathy.[13] Ideally, clinical interventions that are used have demonstrated efficacy in RCTs that account for natural history and the placebo effect of the intervention.

Previous systematic reviews of exercise interventions in mid-portion AT have stated that heterogeneity of outcome measures and populations have led to an inability to provide clear conclusions on the efficacy of HECT.[13] In this review we defined strict inclusion criteria to limit the extent of heterogeneity. While previous reviews have defined the diagnosis to mid-portion AT we also defined the intervention to HECT as defined by Alfredson et al.[6] We also used only one outcome measure for pain and function, the VISA-A, which has recently been suggested as the only valid and reliable tool to assess pain and function in mid-portion AT.[25] Specially, after a pre-planned sensitivity analysis there was minimal heterogeneity in all analyses that included more than one study ($I^2=0\%$). Given the minimal heterogeneity it is important to note that the results of this study may not be applicable to all presentations of mid-portion AT. Specifically, the participants included in this meta-analysis were between the mean ages of 37 to 49, had a mean duration of symptoms varying from 6 to 28 months, and had a mean moderate level of disability on the VISA-A (36–62 points).

HECT has been compared with a natural history group in a single study that had a small sample size (n=25 in each arm) and was at high risk of bias (table 3).[36] Further, no studies have compared HECT to a sham/placebo intervention. While this single study suggests HECT was superior to natural history, the design could not account for the placebo effect in the way that a sham intervention would have. Given the quality of the evidence is very low, as per the GRADE levels of evidence,[38] we are unable to definitively state the effect of HECT compared with natural history. Further, no studies have compared HECT to placebo.

Two studies have compared HECT to traditional physiotherapy, such as ultrasound and tendon friction massage,[16 21] and the pooled mean difference suggested a significant effect in favour of HECT. However, given the quality of the evidence is low, as per the GRADE levels of evidence,[38] we are unable to definitively state the effect of HECT compared with traditional physiotherapy.

Four studies have compared HECT to other exercise interventions[7 10 34 35] and the pooled mean difference suggests no significant difference between the exercise types. These other exercise interventions included heavy slow resistance training, the Stanish intervention and modified formats of HECT. The overall demand on participants with these interventions is significantly lower in terms of quantity of the exercise.[7 10 34] However, given the quality of the body of evidence was very low, as per the GRADE levels of evidence,[38] we are unable to definitively state the effect of HECT compared with other exercise interventions.

As per our intervention, a sensitivity analysis was conducted, and one article was removed from analysis.[35] After removing this article, the pooled mean difference demonstrated a significant effect in favour of other exercise interventions.

It is important to note though that even with one study removed, the pooled mean difference was −5.65 points on the VISA-A. While a minimally clinically important difference (MCID) for the VISA-A is yet to be established, the MCID for the VISA-A has been proposed to be from 10 to 20 in a number of studies.[25] Given a minimum MCID of 10 points, even though the pooled mean difference is significantly in favour of other exercise interventions it is unlikely that this difference is clinically meaningful.

Implications of this systematic review

Quality of the evidence

While the results of this systematic review suggest the use of HECT compared with both natural history and traditional physiotherapy, it is important to consider that the quality of evidence supporting this is very low and low, respectively. This means that there is little to no confidence able to be placed in whether the estimate of the effect represents the true effect.[38] The results of this review also suggest that other exercise interventions may be superior to HECT, the quality of the evidence

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supporting this recommendation is in itself very low. To enable future meta-analysis of quality studies, high-quality, prospective studies with large sample sizes (>200 participants) with either natural history or sham rehabilitation comparators, which report age, BMI, gender, intervention adherence and load monitoring of additional exercise such as running are required.

CONCLUSION

This systematic review included seven studies with participants ranging from a mean (SD) age of 36.6 (7.2) to 49.2 (11.3) years and mean (SD) BMI ranging from 25.0 (5.0) to 31.6 (6.1) kg/m^2. Both males (45%) and females (54%) were included and participants had a duration of symptoms ranging from mean (SD) duration (months) of 6.2 (2.1) to 27.6 (12.9). Therefore, caution must be taken in applying the results of this study to other populations. Nevertheless, three broad conclusions can be drawn from our systematic review.

First, while our study suggests that HECT may be superior to natural history, the quality of evidence is very low and we have little confidence in the estimated effect, as the true effect is likely to be substantially different. Furthermore, no studies have investigated the comparison between HECT and sham rehabilitation.

Second, our study suggests that HECT may be superior to traditional physiotherapy. However, the quality of evidence is low and our confidence in the estimated effect is limited as the true effect may be substantially different from the estimate.

Finally, after a pre-planned sensitivity analysis to remove studies of high risk of bias, our study suggests while HECT may be inferior to other exercise interventions, this difference is unlikely to be clinically meaningful. Likewise, the quality of this body of evidence is low. As such, our confidence in the estimated effect is limited as the true effect may be different.

REPORTING STANDARDS

This systematic review protocol was written as per the PRISMA guidelines and the checklist is attached.

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